# Ashish Kumar

Linkedin: linkedin.com/in/ashish05kumar
Email: meashishkrofficial@gmail.com
GitHub: github.com/Ashishkumar1198
Mobile: +91-9473112005

## Skills

- **Languages:** C/C++, Java, Python
- **Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn
- **Core CS Skills:** Data Structures & Algorithms, OOPS, Time & Space Complexity Analysis
- **Databases:** DBMS, SQL
- **Frontend:** Streamlit, HTML, CSS, Javascript
- **Soft Skills:** Problem-Solving, Collaboration, Analytical Thinking, Adaptability

## Projects

- **India Rainfall Prediction System | Scikit-Learn, Streamlit, Pandas, Matplotlib | Python** (github) Oct '25 – Dec '25
  - Built a machine learning–based rainfall prediction system to forecast next-day rain using historical weather data across multiple Indian cities.
  - Applied Logistic Regression, KNN, and XGBoost models with probability calibration and optimal threshold selection to improve rain/no-rain classification accuracy.
  - Achieved ~85% classification accuracy and ~0.88 ROC-AUC, reduced false rain misses by ~20% through threshold tuning, and delivered an interactive Streamlit dashboard with 6+ analytical visualizations.

- **Social Network Friend Suggestion Tool | C++ | DSA** (github) Aug '25 – Sep '25
  - Deployed a graph-based social network system in C++ using DSA concepts to simulate friend recommendations with mutual connections.
  - Implemented BFS-based shortest path, degree centrality ranking, and efficient user mapping using hash maps, queues, and adjacency lists.
  - Improved proficiency in graph algorithms, BFS/DFS, and optimized data structure usage, aligning with real-world system design and interview requirements.

- **Operating System - File Recovery System | C ,Tkinter | Python** (github) Mar '25 – Apr '25
  - Developed a Windows-based file recovery and disk optimization tool using Python and Tkinter, enabling raw disk imaging, deleted file recovery via shadow copies, and system-level disk maintenance operations.
  - Integrated disk health checks (CHKDSK), disk optimization (defrag), and secure file deletion with administrator-level execution and PowerShell integration for low-level OS control.
  - Reduced manual disk maintenance effort by ~40%, enabled recovery of deleted files in <30 seconds per operation, and consolidated 6+ file system utilities into a single automated GUI application.

## Certificates

- Introduction to Internet of Things | NPTEL (IIT Kharagpur) | (link) Oct '25
- DSA With C++ | SPLEN Technologies | (link) Jul '25
- Crash Course on Python | Coursera | (link) Mar '24

## Achievements

- Solved 230+ DSA problems in LeetCode

## Education

- **Lovely Professional University** Punjab, India
  Bachelor of Technology - Computer Science and Engineering: **CGPA: 7.47** Aug '23 – Present
- **A.N. College** Patna, Bihar
  Intermediate: **Percentage: 71.8** Apr '21 – Mar '22
- **R.P.S. Public School** Patna, Bihar
  Matriculation: **Percentage: 80.8** Apr '19 – Mar '20